SAINTGITS COLLEGE OF ENGINEERING
(Autonomous)

Department of Electronics and Communication Engineering

CAPSTONE PROJECT

FAULT DETECTION IN A TRANSMISSION LINE

Problem B – Group 14

Course: PDEs, Transforms & Optimization
Problem: Transmission Line Signal Integrity
Group: 14

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Academic Year 2025–2026

Abstract

Transmission-line faults can cause signal distortion and affect the reliability of electrical and electronic systems. This project presents a mathematical and computational approach for detecting and locating a hidden fault using synthetic Time-Domain Reflectometry (TDR) data. The transmission line is modelled using the governing equations for voltage and current propagation. The mathematical pipeline involves Partial Differential Equations (PDEs), Laplace transforms, the Finite Difference Method (FDM), and Linear Programming (LP). The LP approach is used to reconstruct the resistance and capacitance profiles along the line. The true fault location is approximately 0.4898 m, while the reconstructed location is approximately 0.4694 m, giving a localization error of one spatial grid cell. The optimization solver completed successfully, demonstrating the potential of the proposed approach for fault localization using simulated data.

Introduction

Transmission lines are used to carry electrical signals from one point to another. Sometimes, faults can occur in these lines due to changes in electrical properties such as resistance and capacitance. Finding the exact location of a fault is important to understand the problem and reduce signal-related issues. In this project, we use mathematical modelling and computer simulation to study the behaviour of a transmission line and identify the approximate location of a fault. The Finite Difference Method (FDM) is used to calculate the changes in voltage and current, while Time-Domain Reflectometry (TDR) is used to study signal reflections. Linear programming is then used to estimate the changes in resistance and capacitance and help locate the fault. Through this project, we aim to understand how mathematical methods and computational techniques can be used to detect faults in transmission lines.

Related Works

1.Tran Hoai and Duong presented a method for detecting and locating faults in transmission lines using Time-Domain Reflectometry (TDR). Their research analysed reflected waveforms to detect faults and estimate fault resistance and load characteristics. The proposed method combined TDR with a wavelet decomposition algorithm. Numerical simulations using MATLAB/Simulink were performed to demonstrate the performance of the method. This research is closely related to this project, which uses mathematical modelling, synthetic TDR data, and linear programming to estimate the approximate location of a fault in a transmission line.

Reference:

Tran Hoai, L., and Duong, A. (2016). Fault Detection on Transmission Lines Using the Time-Domain Reflectometry Method Based on the Analysis of Reflected Waveforms, 2016 IEEE International Conference on Sustainable Energy Technologies (ICSET), October 2016.

2.Besançon, Marcos, Rafael Santos, Didier Georges (2012), in their paper Finite-Difference Modeling Improvement for Fault Detection in Pipelines, investigated the use of finite-difference models for detecting faults in pipeline systems. The researchers improved the numerical model by including instrumentation effects and discussed its application to detecting leaks and instrumentation faults. This work demonstrates how finite-difference modelling can support fault detection and provides a useful mathematical background for this project, which applies numerical simulation and optimization to locate faults in transmission lines.

Reference:

Besançon, G., Guillén, M., Dulhoste, J.-F., Santos, R., and Georges, D. (2012). *Finite-difference modeling improvement for fault detection in pipelines.* IFAC Proceedings Volumes, 45(20), 928–933.

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1 Field Model

Transmission-line behaviour is derived from Maxwell's equations using vector calculus. The relevant parameters per unit length are resistance R, inductance L, conductance G, and capacitance C. For the nominal line,

$$R_0 = 0.1\ \Omega/\mathrm{m}, \quad L_0 = 250\,\mathrm{nH/m}, \quad G_0 = 0, \quad C_0 = 100\,\mathrm{pF/m}.$$

The characteristic impedance is

$$Z_0 = \sqrt{\frac{L_0}{C_0}} \approx 50\ \Omega.$$

Maxwell's curl equation is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Applying the transmission-line approximation gives the telegrapher's equations:

$$\frac{\partial V}{\partial x} = -RI - L\frac{\partial I}{\partial t}, \tag{1}$$

$$\frac{\partial I}{\partial x} = -GV - C\frac{\partial V}{\partial t}. \tag{2}$$

These equations form the physical basis of the transmission-line model.

2 PDE Formulation

Combining the telegrapher's equations gives the damped-wave equations:

$$\frac{\partial^2 V}{\partial x^2} = LC\frac{\partial^2 V}{\partial t^2} + (RC + LG)\frac{\partial V}{\partial t} + RGV, \tag{3}$$

$$\frac{\partial^2 I}{\partial x^2} = LC\frac{\partial^2 I}{\partial t^2} + (RC + LG)\frac{\partial I}{\partial t} + RGI. \tag{4}$$

These are hyperbolic PDEs describing voltage and current propagation with losses. The line length is $\ell = 1\,\mathrm{m}$, with $N_x = 50$ grid points and spacing

$$\Delta x = \frac{\ell}{N_x - 1}.$$

The model uses a synthetic TDR input. A hidden fault near $x = 0.5\,\mathrm{m}$ is introduced by increasing resistance to $4R_0$ and reducing capacitance to $0.6C_0$ over the fault region.

3 Transform Solution

The Laplace transform converts time-dependent equations into the transform domain. Assuming zero initial conditions,

$$\mathcal{L}\left\{\frac{\partial f}{\partial t}\right\} = sF(s).$$

The transformed telegrapher's equations are

$$\frac{\partial V(x,s)}{\partial x} = -(R+Ls)I(x,s), \tag{5}$$

$$\frac{\partial I(x,s)}{\partial x} = -(G+Cs)V(x,s). \tag{6}$$

Differentiating and substituting gives

$$\frac{\partial^2 V}{\partial x^2} = (R+Ls)(G+Cs)V, \tag{7}$$

$$\frac{\partial^2 I}{\partial x^2} = (R+Ls)(G+Cs)I. \tag{8}$$

The propagation constant is

$$\gamma(s) = \sqrt{(R+Ls)(G+Cs)},$$

and the characteristic impedance is

$$Z_0(s) = \sqrt{\frac{R+Ls}{G+Cs}}.$$

For the nominal line,

$$\gamma_0(s) = \sqrt{(R_0+L_0 s)(G_0+C_0 s)}.$$

The propagation constant describes signal attenuation and phase change along the line. The characteristic impedance relates voltage and current waves. The inverse Laplace transform converts the solution back to the time domain, where signal reflections can be interpreted.

These expressions provide a transform-domain description of transmission-line behaviour and a mathematical basis for studying reflected signals. Symbolic tools such as SymPy can be used to represent and simplify the expressions. The project's numerical implementation then uses simulated signals and reconstruction methods to estimate the location of a fault.

4 Discretization

The Finite Difference Method (FDM) approximates the continuous transmission-line equations using discrete spatial and temporal points. For a line of length $\ell = 1\,\text{m}$ with $N_x = 50$ grid points, the spatial step is

$$\Delta x = \frac{\ell}{N_x - 1} = \frac{1}{49} \approx 0.02041\,\text{m}.$$

The first spatial derivative is approximated using the central-difference stencil:

$$\left.\frac{\partial V}{\partial x}\right|_i \approx \frac{V_{i+1} - V_{i-1}}{2\Delta x}.$$

Similarly, the second derivative can be approximated by

$$\left.\frac{\partial^2 V}{\partial x^2}\right|_i \approx \frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta x^2}.$$

These approximations illustrate the FDM stencil, in which neighbouring grid values are used to estimate derivatives. In the implemented forward solver, voltage and current are updated directly over successive time steps rather than by explicitly assembling a global FDM matrix.

The nominal wave velocity is

$$v = \frac{1}{\sqrt{L_0 C_0}} = 2 \times 10^8\,\text{m/s}.$$

The time step is chosen as

$$\Delta t = 0.1\frac{\Delta x}{v} \approx 1.0204 \times 10^{-11}\,\text{s}.$$

Therefore, the Courant–Friedrichs–Lewy (CFL) number is

$$\text{CFL} = \frac{v\Delta t}{\Delta x} = 0.1.$$

The CFL number is 0.1,which satisfies the stability limit selected for the implemented forward solver.Separately, the TDR reconstruction stage combines resistance and capacitance sensitivity matrices to form $Ap \approx b$, which is used in the LP formulation.

5 LP Formulation

Linear Programming (LP) estimates resistance and capacitance variations from the synthetic Time Domain Reflectometry (TDR) response. The reconstruction model is

$$Ap \approx b,$$

where A is the combined sensitivity matrix, p contains the unknown non-negative parameter variations, and b is the synthetic TDR response. The non-negative variables represent resistance increases and capacitance-decrease parameters, allowing both profiles to be reconstructed.

Objective Function: To minimize the reconstruction error while penalizing large parameter variations, the implemented objective is

$$\min_{p,\varepsilon} \left(0.0001 \sum_{j=1}^{n} p_j + \sum_{k=1}^{m} \varepsilon_k \right),$$

where ε_k represents the absolute residual for measurement k.

Constraints and Slack Variables: The absolute residuals are represented by non-negative slack variables subject to

$$-\varepsilon \leq Ap - b \leq \varepsilon,$$

$$p \geq 0, \qquad \varepsilon \geq 0.$$

Equivalently, the residual constraints can be written as

$$Ap - b \leq \varepsilon, \qquad b - Ap \leq \varepsilon.$$

These constraints ensure that each slack variable bounds the magnitude of its corresponding residual.

Inequality Form and Linearity: Define $z = [p^T \ \varepsilon^T]^T$. The LP can be expressed in inequality form as

$$\min_{z} c^T z, \qquad Mz \leq d, \qquad z \geq 0,$$

where c, M, and d represent the objective coefficients and constraint terms.

The problem is linear because the objective and constraints contain only linear combinations of the unknown variables, with no products or powers of unknown variables.

Solution and Convergence: The LP solver successfully find an optimum for the formulated problem.The resulting parameter variations are used to reconstruct the resistance and capacitance profiles and estimate the fault location.

6 Verification

The proposed mathematical pipeline was implemented in Python using Google Colab as the computational environment. NumPy was used for numerical calculations, array operations, and the construction of the transmission-line parameter profiles and sensitivity matrices. Matplotlib was used to visualize the synthetic TDR response, reconstructed resistance and capacitance profiles, and residual error. The "linprog" function from SciPy's optimization module was used to solve the Linear Programming problem and reconstruct the fault-related parameter variations. These tools enabled the numerical implementation of the proposed method and the evaluation of its performance using the simulated data.

The numerical verification results obtained from the Python implementation are summarized below.

The results were verified using two criteria: the reconstructed fault location should be within one spatial grid cell of the actual fault location, and the reconstructed resistance and capacitance values should remain non-negative. Both criteria were satisfied by the simulation results.

No.	Library	Purpose
1	NumPy	Numerical calculations, array operations, and construction of parameter profiles and sensitivity matrices.
2	Matplotlib	Plotting the synthetic TDR response, reconstructed resistance and capacitance profiles, and residual error.
3	SciPy (`linprog`)	Solving the Linear Programming problem to reconstruct fault-related parameter variations.

Verification of the Reconstructed Transmission-Line Parameters

Parameter	Result
True fault location	0.4898 m
Reconstructed fault location	0.4694 m
Grid-location error	1 grid cell (≈ 0.0204 m)
LP solver success	True
Minimum reconstructed R	$0.1\,\Omega/\mathrm{m}$
Minimum reconstructed C	$6.0157 \times 10^{-11}\,\mathrm{F/m}$
Maximum absolute residual error	0.008679 V

7 Pipeline Reflection

The complete mathematical pipeline used in this problem is

$$\boxed{\text{Maxwell} \rightarrow \text{PDE} \rightarrow \text{Laplace} \rightarrow \text{FDM} \rightarrow \text{LP} \rightarrow \text{Verification}}$$

Each stage converts the problem into a form suitable for the next stage.

Six-stage mathematical pipeline

Stage	Math Concept	Course	Equation / Implementation
1	Field Model	Vector Calculus	$\nabla \times E = -\partial_t B$
2	PDE	PDE	Telegrapher equations
3	Transform	Transforms	$\gamma(s) = \sqrt{(R+Ls)(G+Cs)}$
4	Discretization	Numerical Methods	FDM discretization and CFL stability analysis
5	LP	Optimization	$\min \|Ap - b\|_1$ using `linprog`
6	Verification	Numerical Methods, Python Lab	TDR and reconstruction plots

The pipeline demonstrates how a physical transmission-line problem is converted into a mathematical model, transformed, discretized, optimized, and finally verified using numerical results.

8 Results and Discussion

The above techniques were done to detect and locate faults in a transmission line using synthetic Time-Domain Reflectometry (TDR) data. First, the field model was established using the governing electromagnetic equations, which led to the Partial Differential Equations (PDEs) describing voltage and current propagation along the line. The Laplace transform was then used to express the equations in the transform domain, providing a mathematical basis for analysing signal propagation. Next, the Finite Difference Method (FDM) was applied to discretize the governing equations and simulate the transmission-line response. Synthetic TDR data were generated, and a Linear Programming (LP) approach was used to reconstruct the resistance and capacitance profiles along the line. A fault was identified from the deviations of these reconstructed parameters from their normal values. The actual fault location was approximately 0.4898 m, while the reconstructed location was approximately 0.4694 m, corresponding to a localization error of about one grid cell. The optimization solver completed successfully, demonstrating that the proposed mathematical pipeline could estimate the fault location using the simulated data. The following four

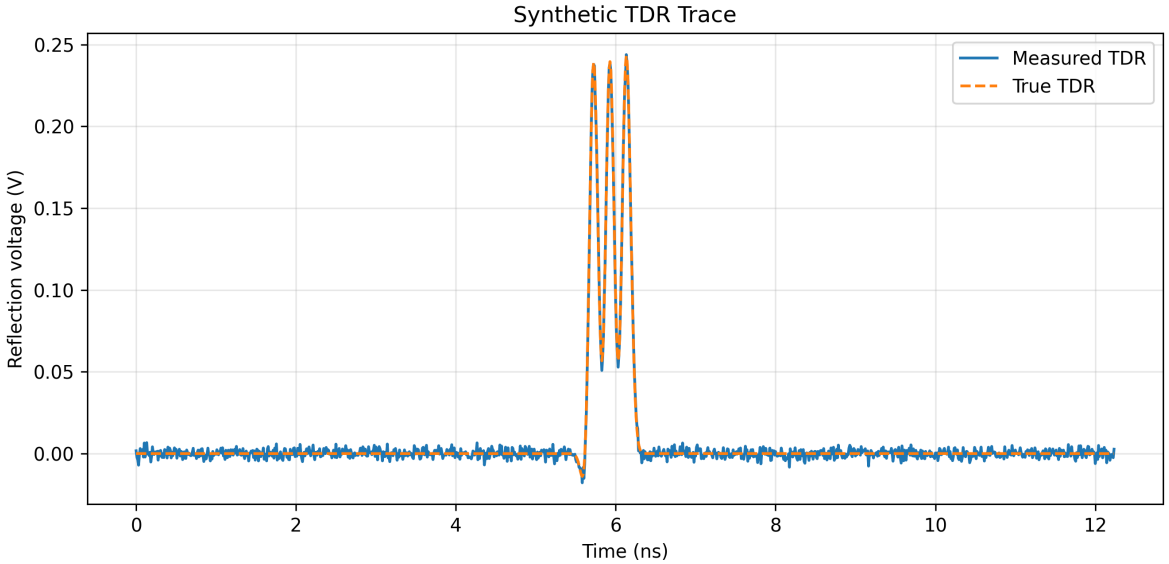
graphs present the synthetic TDR response, reconstructed resistance profile, reconstructed capacitance profile, and residual error, respectively, to evaluate the performance of the proposed method.

8.1 Synthetic TDR Trace

Transmission Line Signal Integrity: Simulation Parameters and Results

Line Parameters	
Length	1.0 m
Number of nodes	50
Spatial step (Δx)	0.02040816326530612 m
Resistance per unit length (R_0)	0.1 Ω/m
Inductance per unit length (L_0)	2.5×10^{-7} H/m
Conductance per unit length (G_0)	0.0 S/m
Capacitance per unit length (C_0)	1×10^{-10} F/m
Characteristic impedance (Z_0)	50.0 Ω
Wave velocity (v)	2×10^{8} m/s
Laplace-Domain Equations	
Propagation constant	$\gamma(s) = \sqrt{(R+Ls)(G+Cs)}$
Characteristic impedance	$Z_0(s) = \sqrt{\frac{R+Ls}{G+Cs}}$
Hidden Fault	
Fault position	0.4897959183673469 m
Fault nodes	23, 24, 25
Fault resistance	0.4 Ω/m
Fault capacitance	6×10^{-11} F/m
FDM Stability	
Time step (Δt)	$1.020408163265306 \times 10^{-11}$ s
CFL number	0.1
CFL condition	Satisfied
FDM simulation	Completed
TDR Data	
Noise level	0.0024315730798507473

The synthetic time-domain reflectometry trace is shown in figure below .



Comparison of the measured and true synthetic TDR signals.

Inference: The graph compares the measured synthetic TDR signal with the true reference signal. A noticeable reflection appears around 5.7–6.3 ns, indicating the presence of a change in the transmission line that produces a reflected signal. The measured signal follows the true signal closely in the reflection region, while small fluctuations are observed due to the added noise. This shows that the simulated TDR response contains information that can

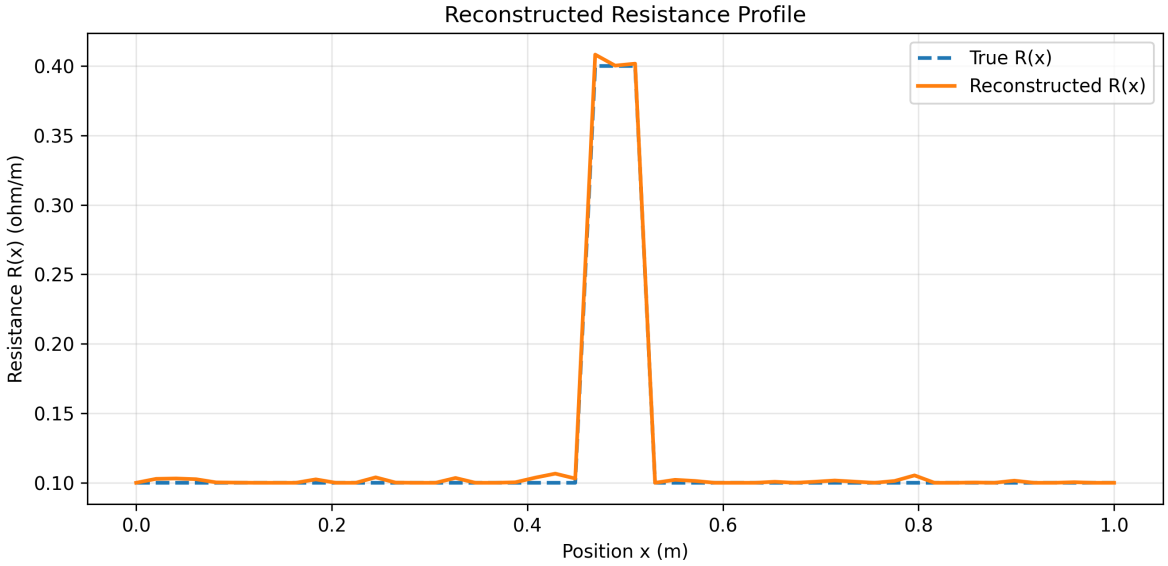
be used for fault detection and further analysis. However, the exact fault location is estimated using the subsequent resistance and capacitance reconstruction process.

8.2 Reconstructed Resistance Profile

The resistance profile results and the comparison of reconstructed and true resistance is given below.

Resistance Profile Reconstruction Results

Parameter	Value
Nominal resistance	$0.1\,\Omega/\mathrm{m}$
Fault resistance	$0.4\,\Omega/\mathrm{m}$
True fault location	$0.4898\,\mathrm{m}$
Recovered fault location	$0.4694\,\mathrm{m}$
Fault location error	1 grid cell ($\approx 0.0204\,\mathrm{m}$)
Minimum reconstructed resistance	$0.1\,\Omega/\mathrm{m}$



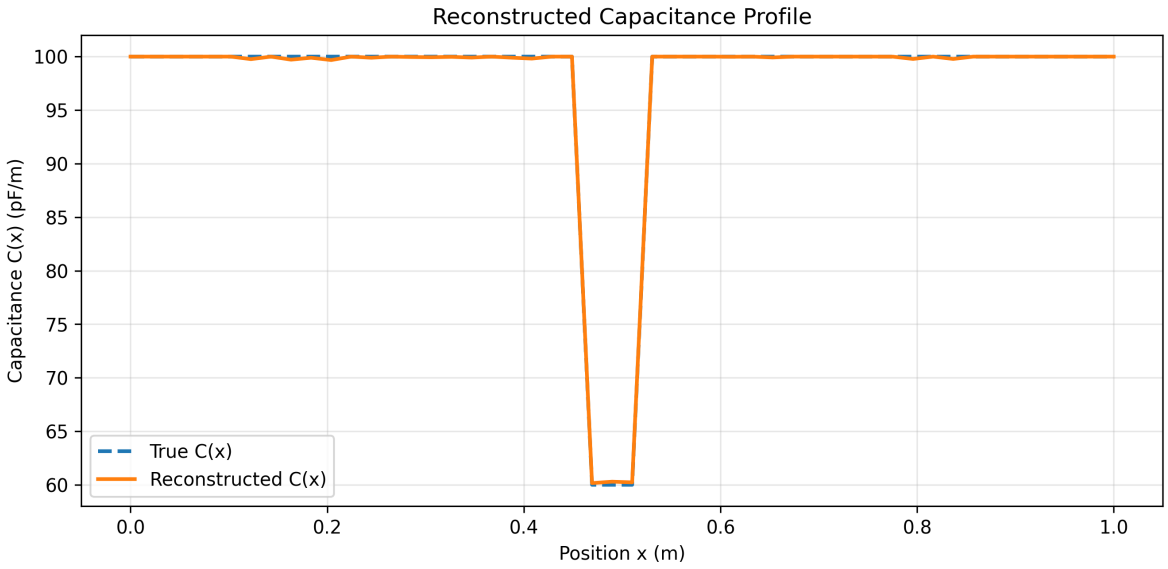
Inference: The reconstructed resistance profile shows an increase in resistance near the fault region compared with the normal transmission line. The normal resistance per unit length is $0.1\,\Omega/\mathrm{m}$, while the fault resistance is increased to $0.4\,\Omega/\mathrm{m}$ in the simulated model. This change helps identify the region where the fault is present. The reconstructed profile estimates the fault location from the variation in resistance along the transmission line. Thus, the resistance reconstruction helps detect and localize the fault.

8.3 Reconstructed Capacitance Profile

The capacitance profile results and the comparison of reconstructed and true capacitance is given below.

Capacitance Profile Reconstruction Results

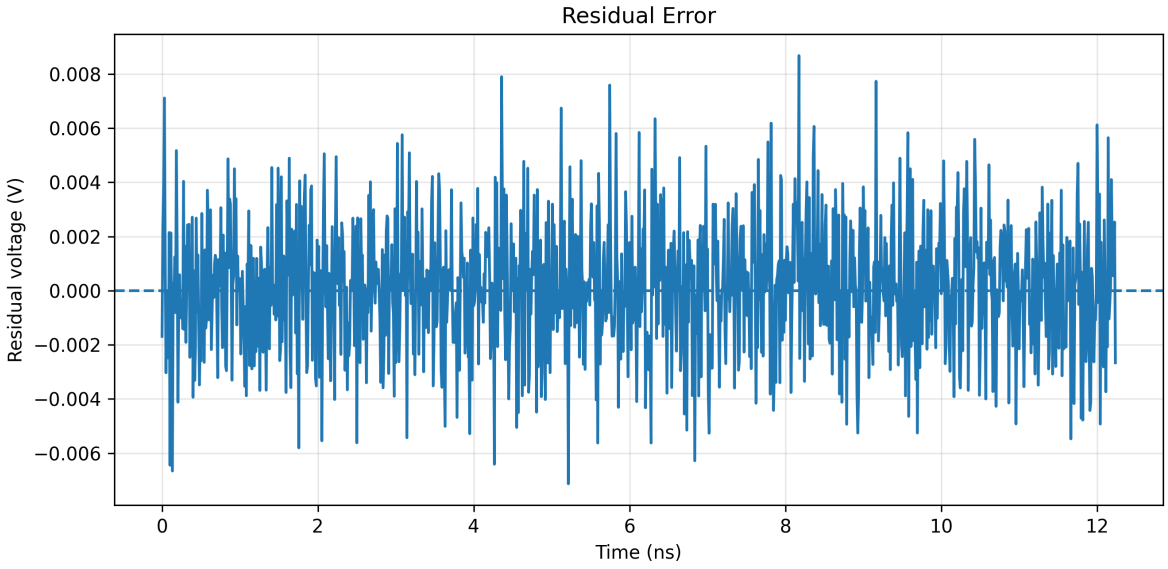
Parameter	Value
Nominal capacitance	$1.0 \times 10^{-10}\,\mathrm{F/m}$
Fault capacitance	$6.0 \times 10^{-11}\,\mathrm{F/m}$
True fault location	0.4898 m
Recovered fault location	0.4694 m
Fault location error	1 grid cell ($\approx 0.0204\,\mathrm{m}$)
Minimum reconstructed capacitance	$6.0157 \times 10^{-11}\,\mathrm{F/m}$



Inference: The reconstructed capacitance profile shows a decrease in capacitance near the fault region. The normal capacitance per unit length is $1 \times 10^{-10}\,\mathrm{F/m}$, whereas the fault capacitance is reduced to $6 \times 10^{-11}\,\mathrm{F/m}$ in the simulated model. This variation helps identify the region affected by the fault. By reconstructing the capacitance distribution along the transmission line, the method provides additional information to support the fault location estimated from the resistance profile.

8.4 Residual Error

The residual error between the model and the reconstructed solution is shown in figure below.



Inference: The residual error graph represents the difference between the model-predicted response and the synthetic TDR data used in the reconstruction process. A small residual error indicates that the reconstructed

model closely matches the given data, whereas a larger residual indicates a greater difference between them. The residual error is used to evaluate the accuracy of the Linear Programming (LP) reconstruction. Thus, this graph helps assess how well the proposed method fits the synthetic TDR data and supports the evaluation of the fault detection process.

9 Limitations

The following limitations are considered to assess the reliability and practical applicability of the proposed technique, including the influence of simulated data, noise, grid resolution, model assumptions, and the absence of real-world testing.

(i) Simulated Data: The study uses simulated TDR data rather than measurements collected from a real transmission line.

(ii) Grid Resolution: The finite spatial grid limits the precision of fault localization.

(iii) Noise Effects: Artificial Gaussian noise may not fully represent noise encountered in practical conditions.

(iv) Model Dependence: Reconstruction accuracy depends on the sensitivity matrices and the Linear Programming formulation.

(v) Real-World Validation: The method has not yet been tested using real transmission-line measurements.

10 Conclusion

This study addressed the problem of detecting and localizing faults in a transmission line by analysing its Time-Domain Reflectometry (TDR) response. A mathematical pipeline was developed by combining the Field model, Partial Differential Equations(PDE), Laplace Transforms(LT), Finite Difference Method (FDM), and Linear Programming (LP) etc. Synthetic TDR data with added noise was used to reconstruct the resistance and capacitance variations along the line. The LP solver successfully reconstructed the fault-related parameter profiles, estimating the fault location as 0.4694 m compared with the actual simulated location of 0.4898 m. The resulting error was one spatial grid cell, demonstrating the potential of the proposed mathematical approach for fault localization using simulated data.

Future Scope

In future, the proposed method can be tested using real transmission-line measurements to evaluate its practical performance. The model can be extended to different fault locations, fault types, and noise conditions. A finer spatial grid and improved sensitivity matrices may help enhance localization accuracy. Further studies can also investigate the method's performance on longer transmission lines and under varying operating conditions to assess its reliability and suitability for practical fault-detection applications.

11 References and Repository

References

1. Tran Hoai and Duong (2016). "Fault Detection on the Transmission Lines Using the Time Domain Reflectometry Method Basing on the Analysis of Reflected Waveform." In *Proceedings of the IEEE International Conference on Sustainable Energy Technologies (ICSET).*

2. LeVeque R.J. (2007). *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems.* Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM).

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4. SciPy Community - *Linear Programming with SciPy:* `scipy.optimize.linprog`. Available at: https://docs.scipy.org/doc/scipy/reference/optimize.linprog-highs.html

5. NumPy Developers - *NumPy Documentation.* Available at: https://numpy.org/doc/

Project Repository

The source code, simulation outputs, and generated plots developed for this project is made available in the following GitHub repository.
GitHub URL: https://github.com/sidharth463/Capstone-Project--GROUP-14